INCH-POUND
MIL-DTL-32068A
11 October 2022
SUPERSEDING
MIL-DTL-32068
12 July 2000

DETAIL SPECIFICATION

LEATHER, CATTLEHIDE, STRAP, VEGETABLE TANNED

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for vegetable tanned cattlehide strap leather.

1.2 Classification. The leather is available in the following types and classes as specified in 6.2:

1.2.1 Types.

Type I - Sides
Type II - Backs
Type III - Bends
Type IV - Double shoulders

1.2.2 Classes.

Class 1 - Mildew resistant treated
Class 2 - Untreated

Comments, suggestions, or questions on this document should be addressed to: Attn: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil/.

AMSC N/A FSC 8330

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. N/A.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Federal Insecticide, Fungicide and Rodenticide (FIFRA)
(40 CFR part 162 State Registration of Pesticides Products)

Federal Insecticide, Fungicide and Rodenticide (FIFRA)
as amended by the Food Quality Protection Act of 1996 and the Pesticide Registration Improvement Act of 2003

(Copies of these documents are available online at https://www.epa.gov/pesticides.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles

(Copies of these document are available on line at https:/aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

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ASTM INTERNATIONAL

ASTM D1813	- Standard Test Method for Measuring Thickness of Leather Test Specimens
ASTM D1814	- Standard Test Method for Measuring Thickness of Leather Units
ASTM D2209	- Standard Test Method for Tensile Strength of Leather
ASTM D2211	- Standard Test Method for Elongation of Leather
ASTM D2617	- Standard Test Method for Total Ash in Leather
ASTM D2810	- Standard Test Method for pH of Leather
ASTM D2813	- Standard Practice for Sampling Leather for Physical and Chemical Tests
ASTM D3495	- Standard Test Method for Hexane Extraction of Leather
ASTM D5053	- Standard Test Method for Colorfastness of Crocking of Leather
ASTM D6018	- Standard Test Method for Determining of Lead Salts in Leather
ASTM D6075	- Standard Test Method for Cracking Resistance of Leather

(Copies of these documents are available online at https://www.astm.org.)

INTERNATIONAL ORGANIZATION for STANDARDIZATION (ISO)

ISO 13365-1 - Leather – Chemical Determination of the Preservative (TCMTB, PCMC, OPP, OIT) Content in Leather by Liquid Chromatography Part 1 – Acetonitrile Extraction Method

(Copies of this document are available online at https://www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, or environmentally preferable or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials. The strap leather shall be full grain or corrected grain cattlehide. Corrected grain leather shall be buffed or snuffed only to the extent necessary to remove superficial blemishes.

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3.3.1 Trim (form and area). The leather shall be furnished in the form of sides, backs, bends or double shoulders as required for the type specified. The area in terms of square feet shall be legibly marked in the butt area on the flesh side of the leather except that double shoulders shall be marked on the flesh side in the right hand corner formed by the straight edge cut and the belly cut.

3.3.2 Tannage. The leather shall be vegetable tanned.

3.4 Color and finish. The color and finish of the leather shall be as specified (see 6.2 and 6.3).

3.5 Physical requirements. The finished leather shall conform to the physical and chemical requirements listed below in Table I, when tested as specified in 4.5. The thickness of the leather shall be as specified in the contract (see 6.2). A thickness tolerance of (± 1/2) ounce (1 ounce = 1/64 inch) is permitted when tested as specified in 4.5.

TABLE I. Physical and chemical requirements.

Characteristic	Requirement
Colorfastness to crocking (min.): Dry Wet	 4.0 3.5
Cracking	No cracking of leather
Hexane soluble material, percent 1/ Min. Max.	 9.0 15.0
Water soluble material, percent (max.) 1/	18
Total ash, percent (max.) 1/	5.0
Acidity (pH) (range)	3.3-5.0
Lead, qualitative	Free of lead

1/ Calculated on a moisture-free basis.

3.5.1 Breaking force. At least 80 percent of the specimens tested shall meet the minimum breaking force values shown in Table II when tested as specified in 4.5.

TABLE II. Breaking force.

Thickness of leather (ounces)	Breaking force (pounds) (min.)
2 to 3-1/2	30.0
3-3/4 to 5-1/2	65.0
5-3/4 and up	110.0

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3.5.2 Elongation. At least 80 percent of the specimens tested shall have elongation values no greater than 20.0 percent at the loads specified in Table III when tested as specified in 4.5.

TABLE III. Elongation.

Thickness of leather (ounces)	Load (pounds)
2 to 3-1/2	20.0
3-3/4 to 5-1/2	45.0
5-3/4 and up	70.0

3.6 Fungicide (mildew resistance), Class 1 only. Unless otherwise specified (see 6.2), the leather shall be treated with one of the following fungicides to provide mildew resistance. The fungicides to treat the leather shall be an EPA registered product that complies with the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) (see 2.2.2).

3.6.1 2-(Thiocyanomethylthio) bensothiazole (TCMTB). The leather shall contain not less than 100 ppm and not more than 400 ppm of active ingredient TCMTB when tested as specified in 4.5.

3.6.2 *Ortho*-phenyl phenol (OPP). The leather shall contain not less than 540 ppm and not more than 2580 ppm of active ingredient OPP when tested as specified in 4.5

3.6.3 Diiodomethyl-*para*-tolylsulfone (DIMPTS). The leather shall contain not less than 300 ppm and not more than 2000 ppm of active ingredient DIMPTS when tested as specified in 4.5.

3.7 Workmanship. The finished leather shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.3 and 4.4 shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5 and 4.6.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits

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(AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination of the end item. The leather for each type shall be examined and defects shall be classified as stated in Table IV. The sample unit for these examinations shall be one (1) side, back, bend or double shoulder as required for each type specified. The lot size shall be expressed in terms of one (1) side, back, bend or double shoulder. The contract or purchase order shall identify the number of samples required for thickness for acceptance and rejection of the leather skins. Separate examinations shall be performed for visual and the thickness, but the same sample may be used for each examination.

TABLE IV. Visual examination defects.

Examination	Defect 1/
Quality of leather	Burn mark on flesh
	Flesh side not smooth or has areas of loose flesh in excess of four (4) square inches in total area.
	Stain, discoloration, cut, scratch, hole, brand, hip mark, rough shoulder, heavy vein, deep fat wrinkle, boney or flanky areas and surface imperfections which impair the cutting value of more than 12 percent of the total area of the leather.
Color (all types)	Not the same color on both sides Not uniform
Area (expressed in terms of square feet)	Not as marked on leather
Marking	Omitted, not readable
Thickness	Not as specified 2/

1/ Light, well healed scratches and grub holes, light fat wrinkles, and slight stains shall not be classified as defects.
2/ Thickness shall be measured in accordance with ASTM D1814

4.5 End item testing. The finished leather shall be tested for the characteristics as specified in Table V. The methods of testing as specified wherever applicable and as listed in Table V shall be followed. Sample grinding and sampling location from which the sample unit is to be obtained shall be in accordance with ASTM D2813. The sample(s) required for testing the lot for physical and chemical characteristics shall be selected from the samples for visual or dimensional thickness examination. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any of the following conditions exists:

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a. More than three (3) failures occur for breaking force or elongation
b. More than one (1) test specimens fail to meet any of the remaining requirements applicable to the sample unit.
c. Any composite fails to meet the specified requirement.

TABLE V. End item tests.

Characteristic	Requirement Reference	Test Method	Requirement	
			Sample unit	Composite sample
Material identification	See 3.3	Visual (see 6.4)		-
Tannage	See 3.3.1	Visual (see 6.4)		-
Color	See 3.4	4.6.1		-
Thickness	See 3.5	ASTM D1813	X	-
Colorfastness to crocking	Table I	ASTM D5053 1/	X	-
Cracking	Table I	ASTM D6075 & 4.6.2	X	-
Hexane-soluble material	Table I	ASTM D3495	-	X
Water-soluble material	Table I	ASTM D3495	-	X
Total ash	Table I	ASTM D2617	-	X
Acidity (pH)	Table I	ASTM D2810	-	X
Lead (qualitative)	Table I	ASTM D6018	-	X
Breaking strength	Table II	ASTM D2209	X	-
Elongation	Table III	ASTM D2211	X	-
Fungicide (Class 1)	See 3.6	ISO 13365-1 2/	-	X

1/ Evaluated using AATCC EP8.
2/ When testing DIMPTS, use an HPLC analytical standard to measure the concentration of DIMPTS in accordance with ISO 13365-1.

4.6 Methods of testing and inspection.

4.6.1 Visual shade examination. The finished leather shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.5) with sources simulating artificial daylight D75 illuminant with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles and shall be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856K (± 200).

4.6.2 Cracking. He leather shall be tested for cracking in accordance with ASTM D6075. The diameters of the mandrels for the indicated thickness shall be as follows:

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Thickness of leather (ounces)	Diameter of mandrel (inches)
2 to 5-1/2	1/4
5-3/4 to 9-1/2	3/8
9-3/4 and over	1/2

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The strap leather is intended primarily for use in post office mail bags, cases, belts, holsters, shoulder straps, and cap visors.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Type and class required (see 1.2)
c. The specific issue of individual documents referenced
d. When first article is required (see 3.1)
e. Color when other than natural color is required (see 3.4)
f. Conformance inspection (see 4.3)
g. Inspection conditions (see 4.3.1)
h. Packaging (see 5.1)

6.3 Standard sample. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

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6.5 Subject term (key word) listing.

Backs
Bends
Flesh
Fungicide
Mildew resistant
Sides

6.6 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL

Preparing Activity:
DLA-CT

Review activities:
Army - AT, MD
Air Force -11

(Project 8330-2022-002)

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